



Question 01 (30 Marks)

Let

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

and define

$$W = \{x \in \mathbb{R}^4 : Ax = 0\}.$$

- (a) Show that W is a subspace of $\mathbb{R}^4$ by showing that $0 \in W$ and verifying closure under vector addition and scalar multiplication.

- (b) Solve

$$Ax = 0$$

and write the general solution in parametric vector form. Hence find a basis for W and determine $\dim(W)$.

- (c) Consider

$$v_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} -1 \\ -1 \\ 1 \\ 1 \end{pmatrix}.$$

Determine whether $\{v_1, v_2, v_3\}$ is linearly independent. If it is dependent, identify one redundant vector.

- (d) Find a basis for the column space $C(A)$ and state its dimension.

- (e) Let

$$c = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Verify that $c \in C(A)$. Hence find all unit vectors $q \in C(A)$ such that $q \cdot c = 0$.

Question 02 (35 Marks)

Let

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}.$$

- (a) Find the characteristic polynomial $\det(A - \lambda I)$ and hence determine all eigenvalues of A , including their algebraic multiplicities. Verify your result using both the trace and determinant relationships.
- (b) Find a basis for each eigenspace. One eigenvalue is repeated; determine the dimension of its eigenspace.
- (c) Determine whether A is diagonalizable. If it is, construct an invertible eigenvector matrix X and a diagonal matrix Λ such that

$$A = X\Lambda X^{-1}.$$

Clearly indicate which column of X corresponds to each diagonal entry of Λ .

- (d) Let

$$\mathbf{u}_0 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}.$$

Express $\mathbf{u}_0$ as a linear combination of eigenvectors of A . Hence obtain a closed-form expression for $A^n \mathbf{u}_0$ for every positive integer n , without computing powers of A by repeated multiplication.

- (e) Suppose

$$B = P^{-1}AP,$$

where P is any invertible 3×3 matrix. Without finding the characteristic polynomial of B , determine the eigenvalues of B , decide whether B is diagonalizable, and find $\det(B)$ and $\text{trace}(B)$. Justify every conclusion using the relationship between similar matrices.

Question 03 (35 Marks)

Let

$$A = \begin{pmatrix} 2 & 0 \\ 1 & 1 \\ 0 & 2 \end{pmatrix}.$$

(a) Compute $A^T A$ and hence determine the singular values of A .

(b) Find orthonormal right singular vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, and use

$$\mathbf{u}_i = \frac{A\mathbf{v}_i}{\sigma_i}$$

to obtain the corresponding left singular vectors.

(c) Using the singular vectors corresponding to the nonzero singular values, construct

$$A = U\Sigma V^T.$$

Clearly state the dimensions of U , Σ , and V .

(d) Using the singular vectors, give orthonormal bases for the row space and column space of A . Hence determine $\text{rank}(A)$.

(e) Write

$$A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T$$

and explain briefly why each term has rank one.